

Project Report

Goldman Sachs (GS) Stock Analysis and Time-Series Forecasting

Exploratory Data Analysis, Data Visualization, Trend Analysis, Volatility & Risk Analysis, and ARIMA-based Forecasting

Dataset: Historical daily OHLCV prices for The Goldman Sachs Group, Inc. (NYSE: GS)

Analysis Period: May 4, 1999 – January 2, 2026

Tools: Python, Pandas, Matplotlib, Seaborn, Statsmodels (ARIMA) — Google Colab

Report Date: August 5, 2026

1. Introduction

The Goldman Sachs Group, Inc. is a premier global investment banking, securities, and investment management firm headquartered in New York City. Founded in 1869, the firm operates across three key business segments: Investment Banking, Global Markets, and Asset & Wealth Management. Goldman Sachs is publicly listed on the New York Stock Exchange under the ticker symbol GS, with a market capitalization in the hundreds of billions of US dollars, making it one of the most influential financial institutions in the world.

This project performs a structured, data-driven analysis of GS's historical stock price behavior. Using a dataset of daily open, high, low, close, adjusted close, and trading volume figures spanning over two and a half decades, the project examines the stock's long-term trend, quantifies its risk and volatility characteristics, and builds a time-series forecasting model to project near-term price movement.

2. Project Objectives

- Perform Exploratory Data Analysis (EDA) to understand the structure, quality, and statistical properties of the dataset.
- Visualize price and volume behavior to identify patterns not obvious from raw numbers.
- Analyze long-term and medium-term price trends using moving averages and yearly aggregates.
- Quantify volatility and downside risk using rolling volatility, Value-at-Risk (VaR), the Sharpe ratio, and maximum drawdown.
- Build and validate a time-series forecasting model (ARIMA) to project future closing prices.

3. Dataset Description

The dataset consists of 6,709 daily trading records for GS common stock, covering the period from May 4, 1999 to January 2, 2026. Each record contains the following fields:

- date — trading date
- open, high, low, close — daily price points in USD
- adj_close — closing price adjusted for dividends and splits
- volume — number of shares traded that day

The dataset was checked for missing values and duplicate records; none were found, indicating a clean, continuous daily series suitable for direct analysis without imputation.

4. Methodology

The analysis was carried out in Python on Google Colab using the following workflow:

- Data loading and cleaning with Pandas (date parsing, chronological sorting, indexing).
- Descriptive statistics and data-quality checks (EDA).
- Visualization with Matplotlib and Seaborn (line charts, histograms, heatmaps).
- Trend analysis using 50-day and 200-day simple moving averages and yearly resampling.

- Risk analysis using daily returns, rolling annualized volatility, historical Value-at-Risk, the Sharpe ratio, and maximum drawdown.
- Forecasting using an ARIMA(5,1,0) model fit with Statsmodels, validated on a held-out 10% test split, then used to project 30 business days forward.

5. Exploratory Data Analysis

Summary statistics show that the closing price ranged from a minimum of \$47.41 to a maximum of \$905.31 across the dataset, with a mean of \$200.99 and a standard deviation of \$136.78 — reflecting substantial price appreciation and variability over the 25+ year period. Average daily trading volume was approximately 4.85 million shares/day, with occasional large spikes coinciding with periods of market stress (e.g., the 2008 financial crisis and the 2020 pandemic sell-off).

Metric	Value
Number of trading days	6,709
Date range	May 4, 1999 to January 2, 2026
Mean closing price	\$200.99
Std. dev. of closing price	\$136.78
Minimum closing price	\$47.41
Maximum closing price	\$905.31
Average daily volume	4.85 million shares/day
Missing values	None
Duplicate records	None

A correlation heatmap of the price and volume fields confirmed that open, high, low, close, and adjusted close are almost perfectly correlated with one another (as expected for OHLC data from the same trading day), while volume shows a weak negative correlation with price levels — consistent with heavier trading activity during the stock's earlier, lower-priced years.

6. Data Visualization

The closing price series below shows the full historical trajectory of GS stock, including the 2008 financial crisis drawdown, the 2020 pandemic dip, and a sharp acceleration in price from 2023 onward.



Figure 1: GS closing price, 1999–2026.

7. Stock Price Trend Analysis

A 50-day and 200-day moving average were overlaid on the closing price to smooth out short-term noise and highlight the underlying trend. Periods where the 50-day average sits above the 200-day average ('golden cross') generally mark sustained bullish phases, while the reverse ('death cross') marks bearish phases.

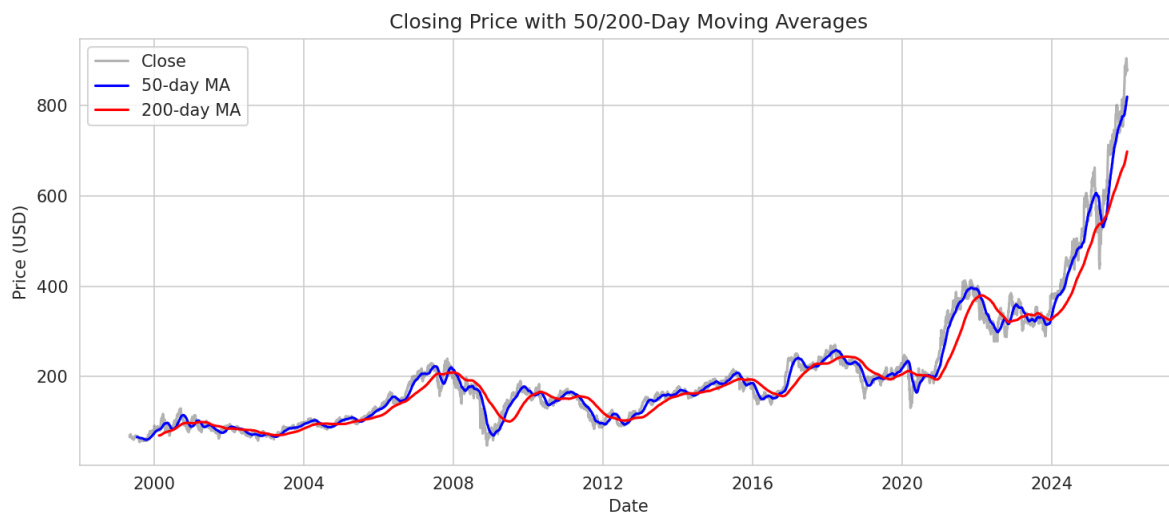


Figure 2: Closing price with 50-day and 200-day moving averages.

Resampling the data to yearly averages makes the long-term trend even clearer: after a long period of range-bound and cyclical prices through the 2010s, the average closing price rose sharply in 2024 and 2025, roughly doubling within two years.



Figure 3: Average closing price by calendar year.

8. Volatility & Risk Analysis

Daily returns were computed from the closing price series and used to derive a set of standard risk metrics, summarized below.

Risk Metric	Value	Interpretation
Historical 1-day VaR (95%)	-2.97%	On 95% of trading days, losses are not expected to exceed this percentage.
Annualized return	15.39%	Average yearly return implied by daily returns.
Annualized volatility	34.37%	Standard deviation of returns, scaled to a yearly basis.
Sharpe ratio	0.45	Return earned per unit of risk (0% risk-free rate assumed).
Maximum drawdown	-80.25%	Largest peak-to-trough decline over the full history.

The 30-day rolling annualized volatility chart shows clear volatility clustering, with sharp spikes during the 2008–2009 financial crisis and the March 2020 pandemic shock, and comparatively calmer periods in between — a well-known characteristic of equity return series.

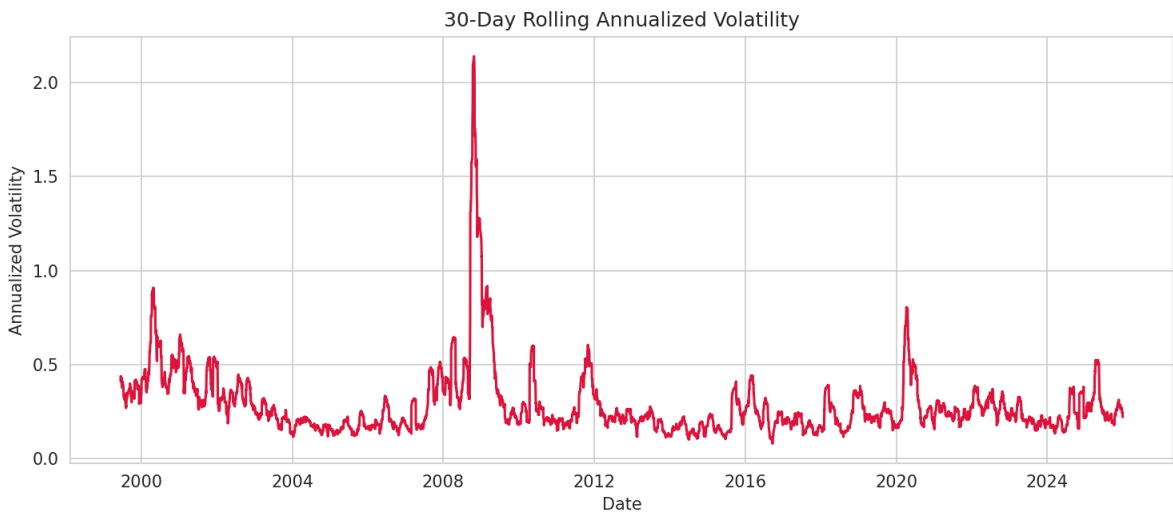


Figure 4: 30-day rolling annualized volatility.

The maximum drawdown of -80.25% indicates that at its worst point, GS stock lost more than four-fifths of its value from a prior peak, underscoring the importance of risk management and diversification for any position sizing decisions involving this stock.

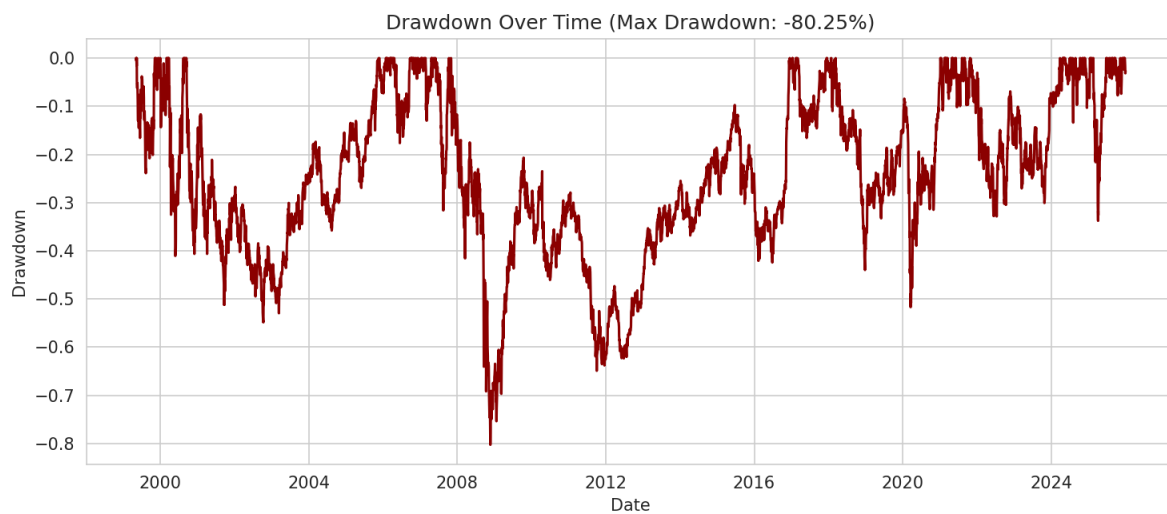


Figure 5: Drawdown from prior peak over time.

9. Time-Series Forecasting Model

An ARIMA(5,1,0) model — five autoregressive terms, first-order differencing, no moving-average terms — was fit on the most recent five years of closing prices, resampled to a business-day frequency. The data was split so that the first 90% trained the model and the final 10% served as a held-out test set for validation.

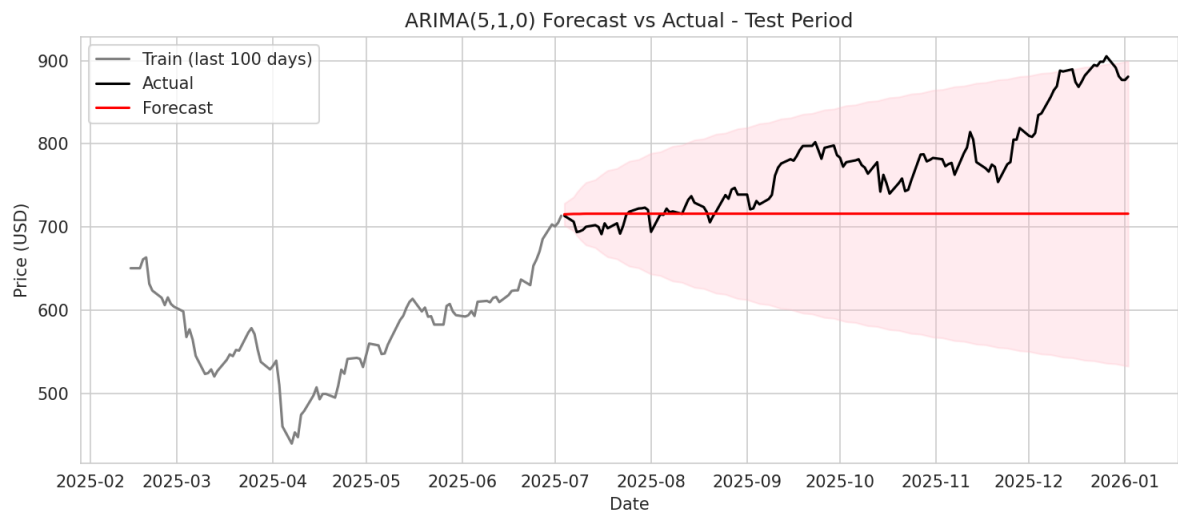


Figure 6: ARIMA forecast vs. actual closing price on the held-out test period.

Forecast accuracy on the test period was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE):

Metric	Value
MAE	\$61.46
RMSE	\$81.31
MAPE	7.51%

A MAPE of 7.51% indicates the model's forecasts were off by roughly that percentage on average during a period when the stock itself moved sharply — a reasonable result for a simple linear model, though not tight enough for precision trading use. The model was then refit on the full 5-year window and used to project 30 business days beyond the last available date.

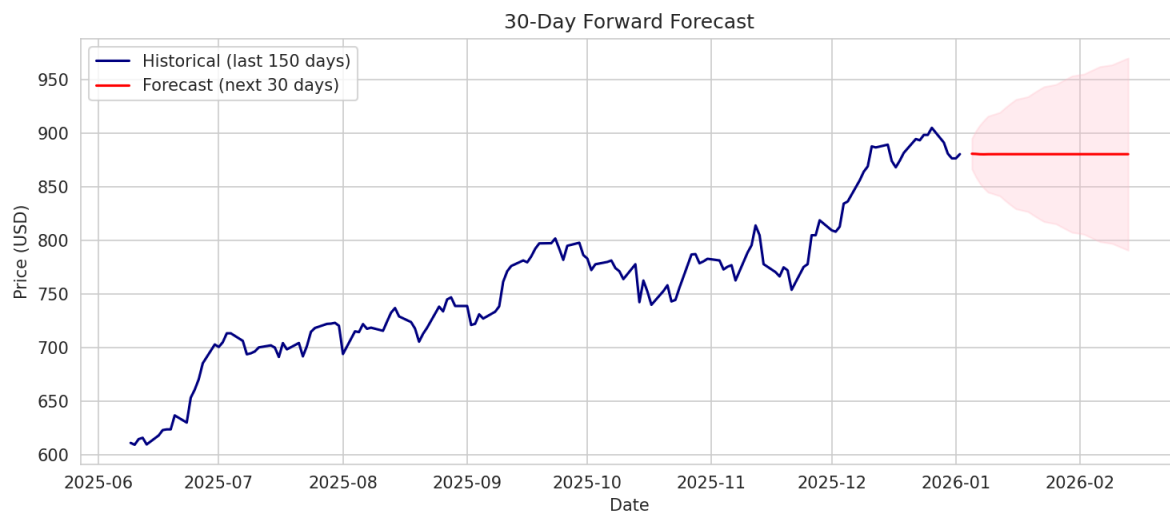


Figure 7: 30-business-day forward forecast with 95% confidence interval.

Starting from a last observed closing price of \$880.75, the model projects a closing price of approximately \$880.70 thirty business days ahead — effectively a flat, near-term projection once the confidence interval is taken into account, reflecting ARIMA's tendency to revert toward a local mean over short forecast horizons rather than extrapolate a strong trend.

10. Conclusion

This analysis of over 25 years of Goldman Sachs stock data shows a company whose share price has been cyclical for most of its history but has appreciated sharply in the last two years, alongside elevated annualized volatility of roughly 34% and a historical maximum drawdown of over 80%. Risk-adjusted returns, as measured by the Sharpe ratio, have been moderate over the long run. The ARIMA(5,1,0) forecasting model captured short-term price dynamics reasonably well on the test set but, like most linear time-series models, tends to produce conservative, mean-reverting forecasts over longer horizons rather than confidently extrapolating recent momentum.

11. Limitations & Future Scope

- ARIMA assumes linear relationships and does not incorporate external factors such as interest rates, earnings announcements, or broader market indices.
- Model order (5,1,0) was chosen as a reasonable default; systematic tuning via ACF/PACF analysis or `auto_arima` could improve fit.
- More advanced models — Facebook Prophet, LSTM/GRU neural networks, or hybrid ARIMA-GARCH models for volatility — could improve both trend and risk forecasts.
- Incorporating macroeconomic indicators, sector indices (e.g., financial sector ETFs), or sentiment data could further improve forecasting accuracy.
- This report and its underlying model are intended for educational and analytical purposes only and do not constitute investment advice.

12. Tools & Technologies Used

- Python 3 — core programming language
- Pandas & NumPy — data loading, cleaning, and numerical computation
- Matplotlib & Seaborn — data visualization
- Statsmodels — ARIMA time-series forecasting
- Scikit-learn — forecast accuracy metrics (MAE, RMSE)
- Google Colab — cloud-based notebook execution environment